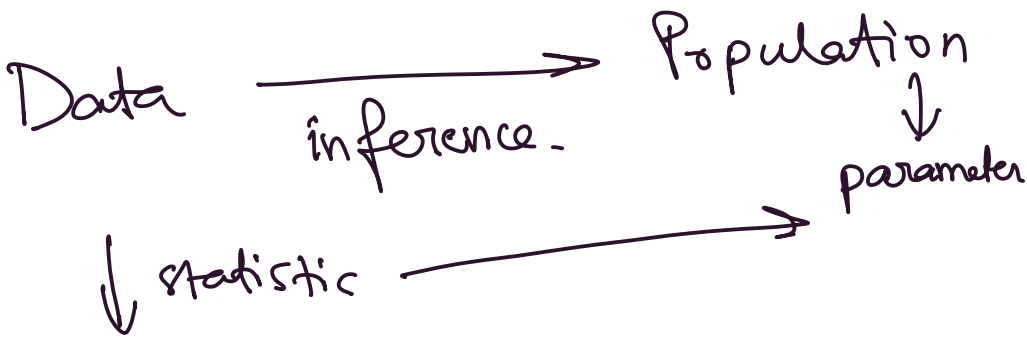
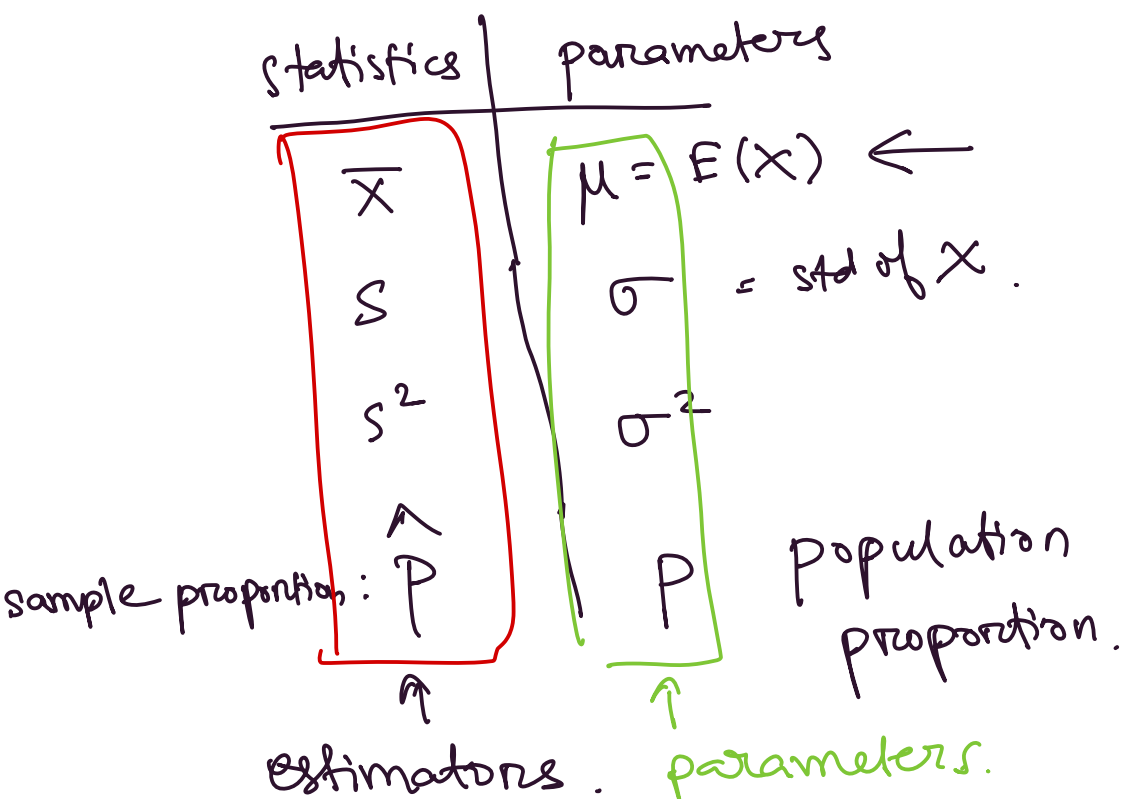
Inferential Statistics.





Estimation.

E(statistic) = parameter.

↓

estimator.

Point Estimates.

(p.e)

What is the p.e of p?

$\hat{p}$ is because $E(\hat{p}) = p$.

$\hat{p}_1, \hat{p}_2, \hat{p}_3, \ldots, \hat{p}_i$

random variable

$E(X) = \mu$ → distribution of X



---

every parameter is associated with the distribution of the point estimate.

CLT :

$$X \sim D(\mu, \sigma)$$

$$\Rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$$X \longrightarrow \bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$X \sim D(\mu, \sigma)$ ← what is the avg grades of a STAT 240 student?

sampling distribution.

$\Rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$

$\bar{X}$ has a dist that is normal $\Rightarrow$ & is centered at μ

$E(\bar{X}) = \mu$

parameter $= \mu$

$\Rightarrow$ p.e $= \bar{X}$

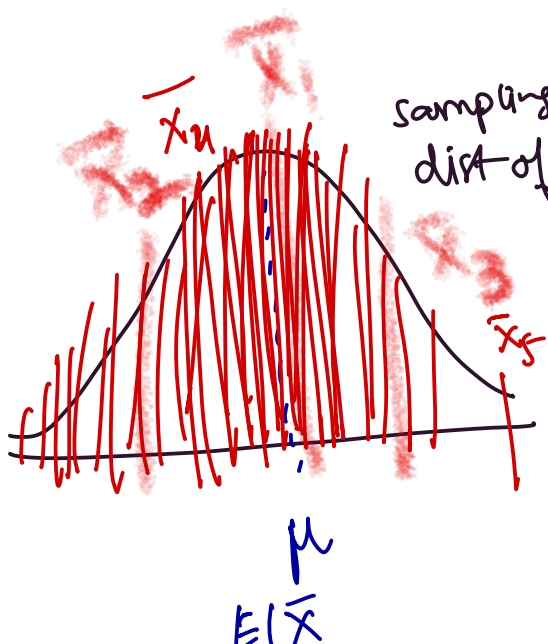
$E(\bar{X}) = \mu$

The dist of a p.e is called a sampling dist.

sampling distribution.

$$\Rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

↳ sampling error or standard error.



dev of this dist is $\frac{\sigma}{\sqrt{n}}$

sampling dist the dev gives you the error in estimating μ.

s.e , p.e , s.d

Lecture Note 6 : Inference

$E(p.e) =$ parameter

p.e $\Rightarrow$ statistic

$\not\Leftarrow$

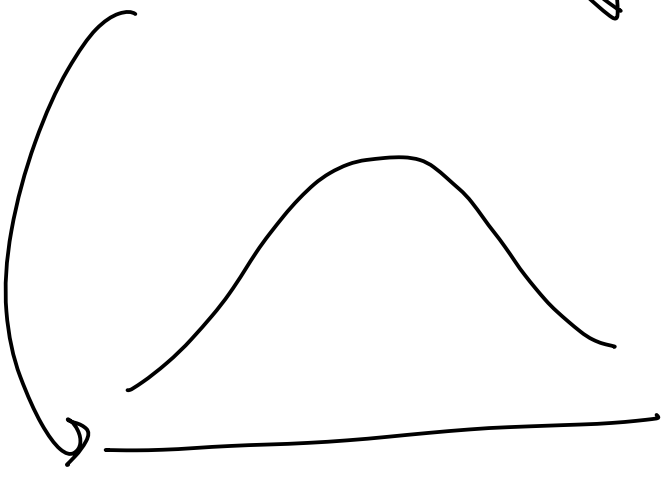
S

parameter : μ.

p.e of $\mu = \overline{X}$

sample mean (statistic)

Sampling dist of p.e.



Std of sampling dist of p.e ?

S.E = standard error or sampling error.

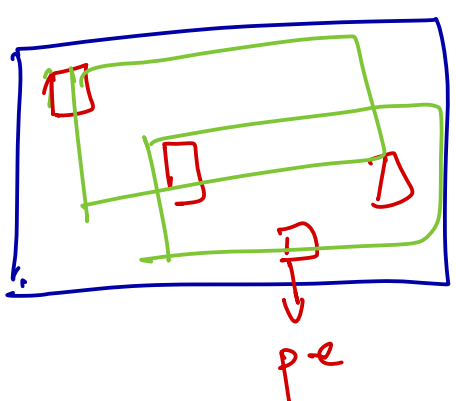
Every s.e of a sampling dist always depends on

n.

where, n = sample size

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$$\downarrow \text{s.e}(\mu) = \frac{\sigma}{\sqrt{n}} \uparrow$$

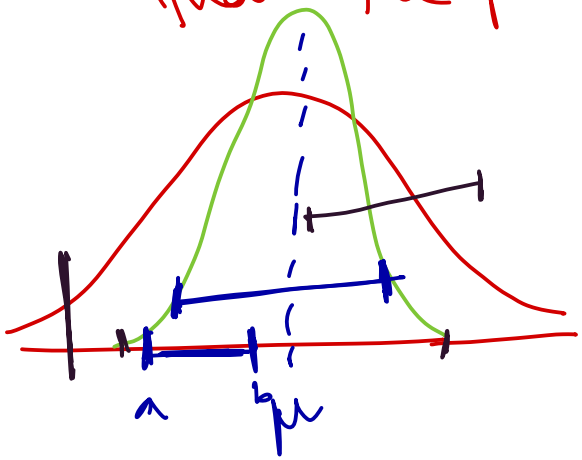


$n = 4$

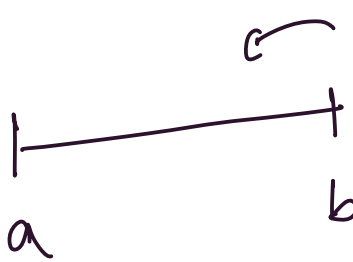
$n = 400$

p·e is closer to the actual parameter

than the p.e of $n = 4$.



s.e $\propto \frac{1}{n}$ ← proportional



Confidence intervals:

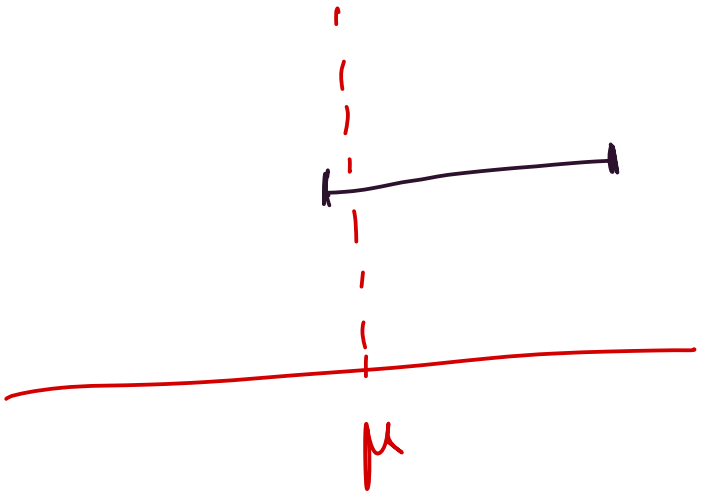
1) give an interval of certain width inside the sampling dist of pe

2) level of confidence. (c)

95% CI of μ is [a, b]

confidence ≠ probability.

confidence ⇒ if you repeat this experiment 100 times, μ will be in the 100c% CI



$c = 0.68$

$\bar{x} + 1d$

$-1d$

$\bar{x} + 2d$

$- 2d$

Confidence Interval.

95 %. CI

c = conf. level = 0.95.

repeat exp. n no. of times & construct the same 100c %. interval every single time

then only 100c %. out n intervals is going to have the actual parameter.

General formula:

$$p.e \pm c.v \times s.e$$

p.e = point estimate

→ c.v = critical value

s.e = standard error.

parameter: μ

$$p.e = \bar{X}$$

$$s.e = \frac{\sigma}{\sqrt{n}}$$

$$c.v = qdist\left(c + \frac{1-c}{2}\right)$$

qdist → sampling dist

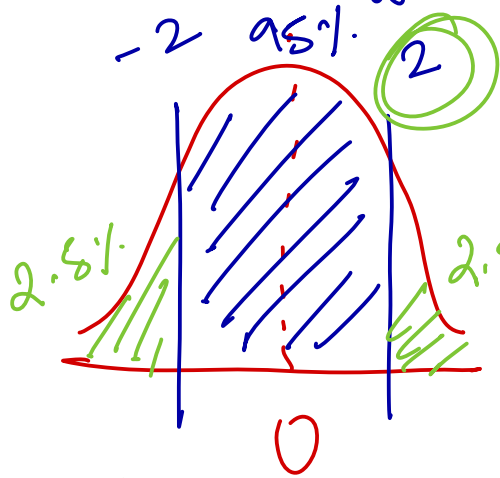
$\frac{1-c}{2}$ → conf. level.

c.v ↓ quantile.

$c + \frac{(1-c)}{2}$ → $0.95 + \frac{(1-0.95)}{2}$

percentile of the (+)ve quantile

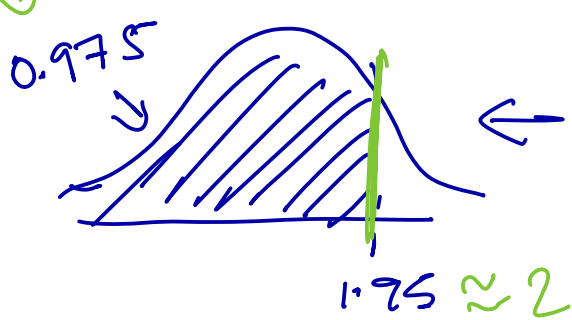
$c = 0.95$



$95 + 2.5$

qnorm $(0.975, 0, 1)$

$= 1.95 \simeq = 2$



z-score of the
97.5th percentile.

And then use

$$x = z\sigma + \mu$$

$$\Rightarrow \text{C.V} \times \text{S.e} \pm \text{p.e}$$

Hypothesis testing:

1) hypothesis about your true parameters value.

H_0: (universal truth) null hypothesis

H_α: (you want) alternative hypothesis.

We will always assume that H_0 is true.

H_0: true parameter $\leq, =, \geq$ value.

H_α: true parameter $>, <, \neq$ value.

↳ direction of H_α

decides the side of test.

H_a: $<$

left tailed test

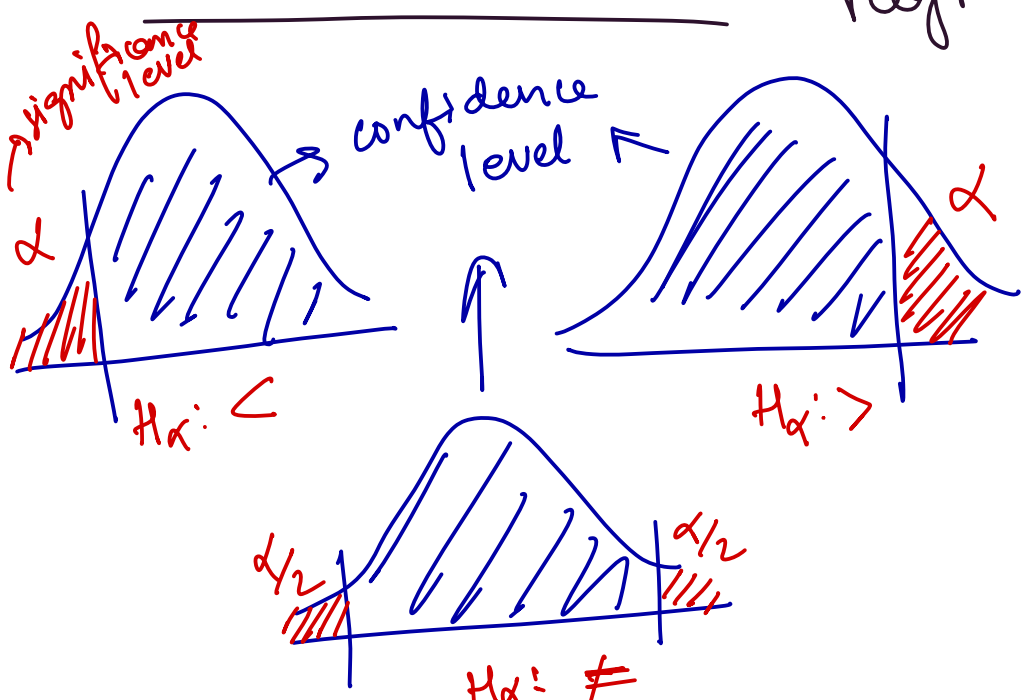
H_a: $>$

right tailed test

H_a: $\neq$

two tailed test.

define the location of your rejection region



p-e → gives evidence for the value of parameter.

↓

$$\text{test statistic} = \frac{\text{p-e} - \text{parameter}_{H_0}}{\text{s.e}}$$

↓ Standardized score.

↓ (format of a z-score)

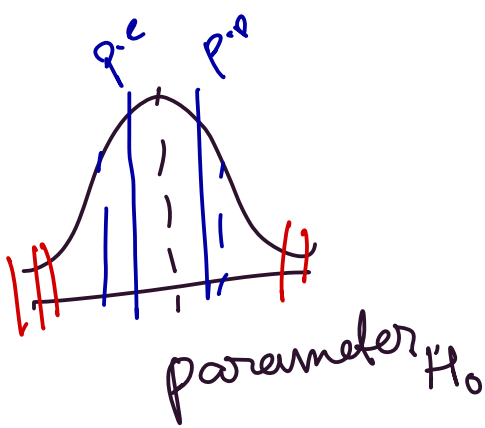
p-e ~ sampling dist.

test statistics ~ standardized version of the sampling dist. = null dist

dist of test statistics is called a null dist.

Next step: to Find the prob. that you got that value of the test statistic. ↓

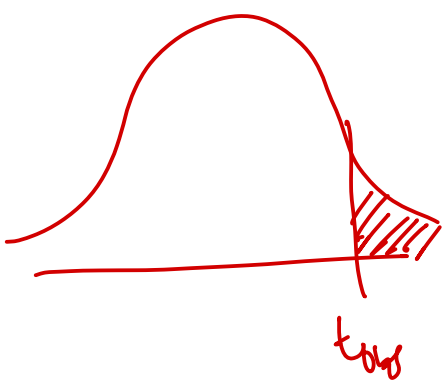
p-value.



p-value = prob of getting the observed value of

the test statistic assuming that the parameters value is the one in H_0.

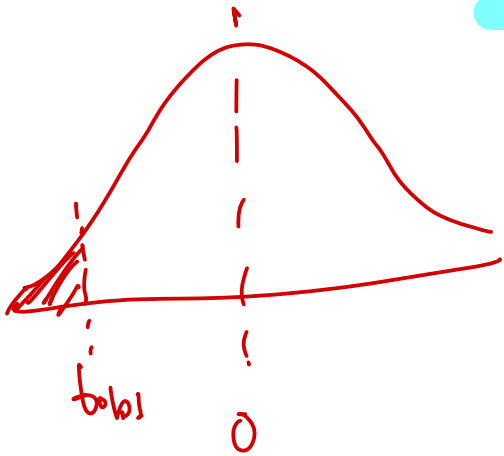
$$p\text{-val} = P(t \geq |t_{obs}|).$$



Left tailed test

$H_a: <$

$P(t \leq -t_{obs})$



Right tailed test

$H_a: >$

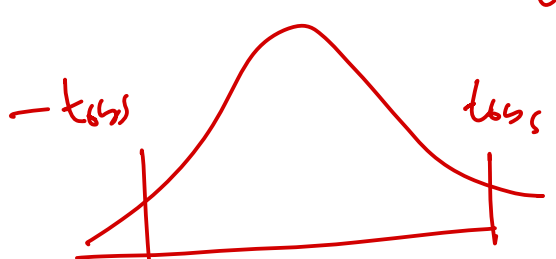
$P(t \geq t_{obs})$



2 tailed test

$H_a: \neq$

$2P(t \leq -t_{obs}$



$2P(t \geq t_{obs})$

Hypothesis testing:

1) H_0, H_α. & understanding

X.

2) test statistics. (evidence from sample)

=

standardized version of p.e

$$= \frac{p.e - \text{parameter}_{H_0}}{s.e(p.e)}$$

$$\left(\frac{p.e - E(p.e)}{s.d(p.e)} \right)$$

~ Null distribution.

(standardized version of the sampling distribution)

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

↙ standardization

$$t \sim N(0,1)$$

3) the prob of getting the observed test statistic.

p-val = Prob of getting a ts as or more extreme than the observed ts.

H_α influences how we calculate the p-val (or how we define extremeness)

$H_\alpha: <$

$P(ts \leq -ts_{obs})$

$H_\alpha: >$

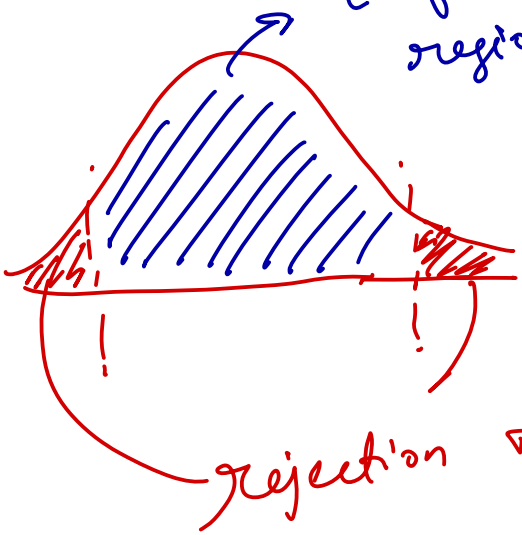
$P(ts \geq ts_{obs})$

$H_\alpha: \neq$ → left tailed test stat.

$p\text{-val} = 2 \times P(ts \leq -ts_{obs})$

$p\text{-val} = 2 \times P(ts \geq ts_{obs})$ → right tailed test stat

$$P(ts \geq |ts_{obs}|)$$



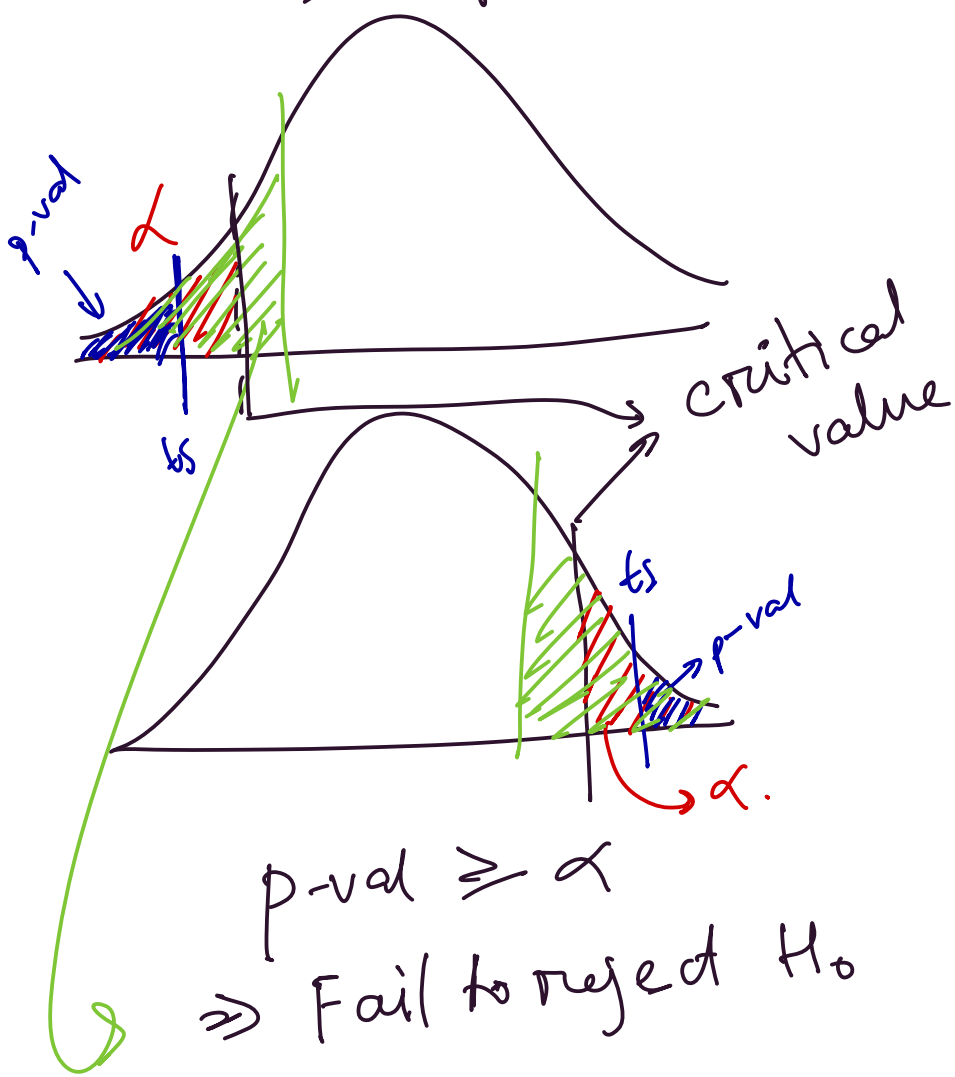
$$\alpha = 1 - c$$

$$c = 1 - \alpha.$$

4) Conclusion of the test

p-val $<$ α

$\Rightarrow$ Reject H_0.



p-val $\geq$ α

$\Rightarrow$ Fail to reject H_0

Even if all the evidence comes from the sample (for or against H_0) if all is uncertain.